		Target engagement	Dose	Efficacy	Safety
#15	**hepatitis C** HCV NS5A inhibitors (10)	**2011** HCV subgenomic replicon (Huh-7 cells; genotype 1a/1b and patient-derived NS5A hybrid…	**2015** HCV subgenomic replicon (Huh-7 cells; genotype 1a/1b and patient-derived NS5A hybrid…	**2020** HCV viral-kinetic mathematical model (Perelson/Dahari lineage) fitted to…	**2015** Repeat-dose toxicology in… Rat/monkey adrenal signal absent in patients
	hepatitis C HCV NS5B (non-nucleoside) inhibito…	**2014** HCV genotype 1 subgenomic replicon (Huh-7), with and without human serum	**2014** Rat and dog repeat-dose toxicology (dasabuvir)	**2018** Human-hepatocyte chimeric (uPA/SCID) mouse infected with… ■ Filibuvir add-on failed SVR desp…	**2014** Rat and dog repeat-dose toxicology (dasabuvir)
	hepatitis C HCV NS5B (nucleotide) inhibitors (7)	**2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2013** Rat, dog and mouse repeat-dose toxicology (GS-9851 / sofosbuvir IND package)	**2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2016** Rat, dog and mouse… Tox missed BMS-986094 heart failure
#16	**diabetic kidney disease** NR3C2 antagonists (3)	**2021** Rat urinary Na+/K+ natriuresis under MR-agonist challenge	**2021** Phase 2b population PK/exposure-response model (ARTS-DN, ARTS-DN Japan)…	**2020** Uninephrectomised DOCA-salt rat	**2021** Rat whole-body autoradiography and equinatriureti… ■ Rat 'no hyperkalaemia' failed in…
#17	**acute myeloid leukaemia** FLT3 inhibitors (4)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2012** Ba/F3 FLT3-ITD resistance screen ■ Combination: no survival gain	**2018** Rat and dog toxicology and safety pharmacology PRES and QT missed by tox species
	acute myeloid leukaemia CD33-directed depleters (3)	**2021** Primary AML blasts + NK cel… NK activation unchanged despite CD33 saturation	**2018** CD33+ AML cell-line and… Phase 3 regimen failed, program abandoned	**2025** Cord-blood CD33-CAR-NK (4-1BB, sIL-15) in NSG Molm13-lu… ■ No single-agent responses in R/…	**2025** Cord-blood CD33-CAR-NK (4-1BB, sIL-15) in NSG Molm13-luc xenografts + HSC colony assay
#18	**malignant pancreatic neoplasm** MAP2K1/2 inhibitors (3)	**2012** Tumour xenografts, pERK by IHC (trametinib)	**2013** Pancreatic cell-line and… Target reached, no benefit, programme dropped	**2018** In-vitro cancer cell-line pane… KRAS status did not pick pimasertib responders	**2013** Toxicology in rat and dog… Animals missed ocular harm, LVEF late
#19	**MASH** NR1H4 agonists (4)	**2019** Rodent FXR target-gene PD (tropifexor)	**2019** STAM and AMLN-diet NASH mice (tropifexor)	**2021** STAM and AMLN-diet NASH mice (tropifexor) □ Outcome trial negative	**2024** High-fat-diet mouse ALT/cholesterol screen ■ Cilofexor: no lipid change
	MASH FGF21 analogue (FGFR1c/KLB)…	**2023** Obese/diabetic cynomolgus monkeys	**2023** Pegozafermin population PK/PD exposure-response model… ■ Reached, failed fibrosis endpoin…	**2023** Pegozafermin population PK/PD exposure-response model… ■ Pegbelfermin failed fibrosis…	**2020** Diet-induced obese mice given recombinant human FGF21… ■ Mouse bone loss not seen at 16…
#20	**prostate cancer** AR antagonists (5)	**2013** LNCaP/AR castrate xenografts	**2013** LNCaP/AR castrate xenografts	**2016** LNCaP/AR castrate xenografts	**2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide)
	prostate cancer PARP1/2 inhibitors (5)	**2009** BRCA2-null synthetic lethality: BRCA2-deficient V-C8 and ES cells and V-C8 xenografts killed by PAR…	**2015** BRCA2-null synthetic lethality: BRCA2-deficient V-C8 and ES cell… ■ Exposure reached; HRR-negativ…	**2023** BRCA2-null synthetic lethality: BRCA2-deficient V-C8 and ES cell… ■ HRR-negative cohort stopped fo…	**2023** Rat and dog repeat-dose toxicology of olaparib, niraparib an… ■ Anaemia, neutropenia unforese…
	prostate cancer CYP17A1/AR inhibitors (3)	**2018** VCaP xenografts in mice plus cynomolgus monkey and rat steroidogenesis	**2018** VCaP xenografts in mice pl… Exposure plateaued; PK stopped development	**2015** LAPC4 androgen-dependent xenografts in SCID mice (galetero… ■ 1/17 PSA50 after enzalutamide;…	**2018** VCaP xenografts in mice plus cynomolgus monkey and rat… ■ CNS toxicity closed the prostate…
	prostate cancer EZH2 inhibitors (3)	**2019** Human CRPC cell lines and xenografts (EZH2 inhibitor + AR blockade)	**2026** Human CRPC cell lines and xenografts (EZH2 inhibitor + AR blockade)	**2026** Human CRPC cell lines and xenografts (EZH2 inhibitor + AR blockade)	**2026** Rat and monkey repeat-dose toxicology (tazemetostat)
	prostate cancer FOLH1-directed depleters (3)	**2017** Docetaxel-refractory PSMA-high CRPC xenografts dosed with PSMA ADC	**2011** Docetaxel-refractory PSMA-high CRPC xenografts dosed with PSMA ADC	**2022** PSMA-positive vs PSMA-negative prostate tumour-bearing mice	**2022** GLP single-dose rat and minipig and 4-week repeat-dose ra… ■ Rodent salivary PSMA lowest;…
#21	**follicular lymphoma** CD19-directed depleters (3)	**2011** Syngeneic BALB/c A20 lymphoma, first-generation anti-mouse-CD19 CD3zeta…	**2015** Raji CD19+ Burkitt xenograft in NSG mice, FMC63-4-1BB CAR T cells (liso-cel lineage)	**2024** Raji CD19+ Burkitt xenograft in NSG mice, FMC63-4-1BB CAR T cells (liso-cel lineage)	**2023** Humanized (HuSGM3) NSG mice with high B-ALL burden, CD1… ■ Lenzilumab did not prevent…
	follicular lymphoma PIK3CD inhibitors (3)	**2011** Patient B-cell tumour samples and B-cell lines ex vivo	**2024** Healthy-volunteer whole-blood basophil activation test (ex-vivo PI3Kδ PD assay for ME-4…	**2014** Patient B-cell tumour samples and B-cell lines ex vivo	**2014** Rat and dog repeat-dose toxicology
#22	**ovarian cancer** PARP1/2 inhibitors (6)	**2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib)	**2021** NOVA exposure–safety model (niraparib)	**2016** BRCA-deficient cancer cells and xenografts (niraparib)	**2017** Dog cardiovascular safety pharmacology (niraparib)
#23	**rheumatoid arthritis** BTK inhibitors (8)	**2013** BTK occupancy probe in CIA mice	**2022** CIA occupancy-efficacy PK/PD Target reached, no effect	**2019** Collagen-induced arthritis (rodent) CIA-active drug failed Phase 2	**2021** Human platelet aggregometry and humanized-mouse thrombus formation (acalabrutinib vs ibrutinib)
	rheumatoid arthritis SYK inhibitors (3)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) ■ Target reached, DAS28 unmove…	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) ■ DAS28 no better than placebo	**2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib) ■ Serious infections missed,…
	rheumatoid arthritis anti-CSF2 antibodies (3)	**2018** Mouse GM-CSF-IRF4-CCL17 pathway in zymosan/antigen arthritis pain models	**2025** Cynomolgus monkey 13-week SC and 26-week IV GLP… ■ Weekly dosing reached; failed in…	**2023** Mouse collagen-induced an… Failed in targeted-therapy-IR; met ACR20 elsewhere	**2025** Human anti-GM-CSF-autoantibody… ■ No PAP in 2915 patients over 2.…
#24	**multiple myeloma** TNFRSF17-directed depleters (8)	**2016** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2014** BCMA+ myeloma xenografts in mice, ADC	**2019** bb2121 CAR T cytotoxicity against cell lines of quantified BCMA density (in vitro)	**2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin)
	multiple myeloma PSMB5 inhibitors (3)	**2015** Human MM.1S myeloma xenografts in CB-17 SCID mice; tumour 20S beta5 inhibition +…	**2015** Ixazomib population-PK model (NONMEM, 226 phase 1… ■ Dosed to MTD, failed, abandoned	**2015** Human MM.1S myeloma xenografts in CB-17 SCID mice;… ■ 9% PR against marked PDX…	**2015** Ixazomib GLP toxicology: 6-month rat and 9-month dog oral… ■ Tolerated in dogs; GI toxicity…
#25	**melanoma** BRAF inhibitors (3)	**2010** BRAF V600E cell lines and xenografts	**2011** Colo205 BRAF V600E xenograft (exposure-response)	**2010** BRAF V600E vs wild-type cell-line panel (patient selection) ■ INTERIM: intermittent arm inferi…	**2014** HRAS Q61L keratinocytes and DMBA/TPA mouse skin… ■ FDA table: cuSCC ~29%, anima…
	melanoma anti-PDCD1 antibodies (3)	**2021** Humanised-PD-1 C57BL/6 mouse with S180 homografts, 124I-toripalimab immuno-PET	**2019** Human T-cell PD-1 occupancy in vitro (saturation threshold)	**2026** Human PD-1 knock-in mouse bearing MC38 tumours	**2024** Cynomolgus monkey… Endocrine and skin irAEs in melanoma cohort
#26	**Crohn disease** anti-IL23A antibodies (3)	**2018** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb)	**2022** Sponsor population-PK / exposure-response dose model from phase 2 (risankizumab; mirikizuma…	**2017** T-cell transfer colitis mouse (anti-mouse IL-23p19 mAb)	**2022** Cynomolgus monkey… Hepatotoxicity in Crohn's; monkeys silent
#27	**open-angle glaucoma** ROCK inhibitors (3)	**2018** Living mouse eye, OCT of conventional outflow tissue, topical netarsudil	**2015** Normotensive monkey eye, topical netarsudil 0.04%	**2015** Normotensive monkey eye, topical netarsudil 0.04% ■ Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
Allergy	**allergic disease** allergen (immunotherapy) (5)	**2019** Patient PBMC Fel d 1-specific CD4+ T cells (ex vivo), Cat-PAD	**2016** Allergen exposure chamber… Chamber regimen reached in field, failed	**2018** Mouse allergic airways… Mouse protection, field loss to placebo	**2019** Allergen exposure chamber (EEC)
	allergic rhinitis PTGDR2 antagonists (4)	**2013** Human whole-blood PGD2 eosinophil shape-change assay	**2017** Bronchial allergen challeng… Challenge regimen reached, failed, abandoned	**2017** Bronchial allergen challeng… PoM-backed regimen failed Phase 3 primary	**2017** Bronchial allergen challenge in mild allergic asthmatics (setipiprant proof-of-mechanism)
	allergic rhinitis allergen (immunotherapy) (8)	**2018** Conjunctival provocation test (CPT), gp-ASIT+	**2019** Conjunctival provocation te… CPT dose reached, confirmatory trial missed	**2019** Conjunctival provocation te… Confirmatory field trial missed primary	**2019** Allergen exposure chamber (EEC)
Autoimmune	**psoriasis** JAK1 inhibitors (3)	**2018** Human whole-blood cytokine-induced STAT phosphorylation (JAK1 potency an…	**2022** Rat (6-month) and cynomolgus monkey (39-week) repeat-dose toxicology	**2018** Rat adjuvant-induced arthritis (cross-indication; no psoriasis model was run)	**2022** Rat (6-month) and… Dose-related platelet fall the animals missed
	psoriasis PDE4 inhibitors (3)	**2020** Human Th17-driven ex vivo skin inflammation model (InflammaSkin; LEO 29102 cream…	**2011** Mouse phorbol-ester ear… Regimen reached, Phase-3-design trial failed	**2019** In-vitro PDE4 enzyme and cellular assays with skin-delivery… ■ Imiquimod mouse worked;…	**2011** Rabbit primary skin irritation and mouse local lymph node assay (2% ointment)
	psoriasis RORC inverse agonists (4)	**2021** Human whole-blood IL-17 secretion assay ■ No IL-17 change in plaques	**2021** Human whole-blood IL-17 secretion assay	**2022** IL-23 minicircle mouse, oral BI 730357	**2015** Nonclinical toxicology species (not public)
	psoriasis anti-IL17A antibodies (4)	**2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** Cynomolgus repeat-dose toxicology	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** People with inborn errors of IL-17 immunity